

Unix Review

Navigating directories, copying files, using virtual machines, and more are a regular part of a machine learning/data science practitioner's job. You will often find the Unix Shell utilized to accomplish these tasks.

Objectives

- What is Unix
- Uses of Unix Shell
- Unix Resources/Exercises

What is Unix

Unix is a powerful, multi-user, and multitasking operating system originally developed in the 1960s and 1970s at AT&T's Bell Labs. It has since evolved into various derivatives and flavors, like Linux and BSD, which share many of the core Unix principles and functionalities. Unix and its variants are widely used in servers, desktops, and embedded systems due to their stability, efficiency, and extensive networking capabilities.

Uses of Unix Shell

1.Many Cloud Computing Platforms are Linux based (utilize a flavor of Unix Shell). For instance, if you want to setup a data science environment on any cloud platform, it really helps to have some Unix Shell knowledge. There are times when you may have a use for a Windows virtual machine, but it is less common. Unix excels in networking capabilities, which is essential for accessing and processing data stored on remote servers, using cloud computing resources, or collaborating with others in distributed systems.

2.Unix Shell provides a number of useful commands such as: `wc` command which counts the number of lines or words in a file, `cat` command which concatenates/merges files, `head` and `tail` commands which help you subset large files, and more.

3. You will often find Unix Shell integrated with other technologies. For example, it is common to find shell commands in Jupyter Notebooks alongside Python code. In Jupyter Notebook, you can access shell commands by escaping to the shell by using an `!`. In the code below, the result of the shell command `ls` (which lists all the files in the current directory) is assigned to the Python variable `myfiles`.

```
myfiles = !ls
```

The image below shows some Python Code integrated in a workflow to combine multiple datasets. Notice a Unix Shell command (enclosed in the red rectangle) integrated in a Jupyter Notebook.

```
import pandas as pd
```

```
# Assign all files ending in .csv in working directory to Python variable myfiles
myfiles = !ls *.csv
```

```
# csv filenames that are in my current directory
myfiles
```

```
['US_Population_1901_1910.csv',
 'US_Population_1911_1920.csv',
 'US_Population_1921_1930.csv',
 'US_Population_1931_1940.csv',
 'US_Population_1941_1950.csv',
 'US_Population_1951_1960.csv',
 'US_Population_1961_1970.csv',
 'US_Population_1971_1980.csv',
 'US_Population_1981_1990.csv']
```

```
finaldf = pd.DataFrame()
```

```
for onefile in myfiles:
    df = pd.read_csv(onefile)
    finaldf = finaldf.append(df)
```

```
finaldf.head()
```

	year	AK	AL	AR	AZ	CA	CO	CT	DC	DE	...
0	1901.0	0.0	1907000.0	1341000.0	131000.0	1550000.0	581000.0	931000.0	285000.0	187000.0	...
1	1902.0	0.0	1935000.0	1360000.0	138000.0	1623000.0	621000.0	952000.0	290000.0	188000.0	...
2	1903.0	0.0	1957000.0	1384000.0	144000.0	1702000.0	652000.0	972000.0	295000.0	190000.0	...
3	1904.0	0.0	1978000.0	1419000.0	151000.0	1792000.0	659000.0	987000.0	302000.0	192000.0	...
4	1905.0	0.0	2012000.0	1447000.0	158000.0	1893000.0	680000.0	1010000.0	308000.0	194000.0	...

5 rows × 58 columns

Keep in mind that the code in the image above isn't some unique way to do a task, but just a small example of how you may see Unix commands utilized.

Unix Resources/Exercises

Jupyter Notebooks and Datasets

Download and unzip this zip file: [Week0_UnixBasics.zip](#)

(<https://vetsintech.instructure.com/courses/241/files/45311?wrap=1>)_ [↓](#)

(https://vetsintech.instructure.com/courses/241/files/45311/download?download_frd=1)

We will go over the contents of the zip file in lecture.

1. Unix_Sys_And_OS.ipynb
2. Unix_Exercises.ipynb

Throughout this course, we will continuously use some basic Unix commands so don't worry if you don't understand all these commands.

→ (<https://youtu.be/6iF8Xb7Z3wQ?si=iLVNFEwb4iUditdq>)

→ (<https://youtu.be/6iF8Xb7Z3wQ?si=iLVNFEwb4iUditdq>)